

Foundational mathematical language: some exercises

Work through this sheet as much as you can stomach. It asks you to prove some foundational facts about sets and functions. This material will appear repeatedly throughout your mathematical studies, but will rarely, if ever, be taught on its own in class. Rather, it will be used routinely without explanation, so it is important to be thoroughly familiar with it.

At the same time, this material is “only language” and, to some of you at least, may feel completely disconnected from your idea of mathematics. Also, formulating the required precise proofs may feel strangely difficult at this point (but it will also provide good practice in mathematical writing). So engage with this material as your spirit allows right now, and return to it when you are ready and when it is needed. I assure you that it will be, so please persist. At some point the results of exercises 1 and 2 should enter your bloodstream (they are also short work if you do them right) and all parts of exercise 3 should become easy enough to do effortlessly whenever required.

Feel free to start with any exercise that looks approachable, e.g. 3, and bounce around as you see fit.

1. A function $f : S \rightarrow T$ is called *invertible* or an *isomorphism of sets* if there is a function $g : T \rightarrow S$, called an inverse of f , such that $f \circ g = id_T$ and $g \circ f = id_S$. Show that (a) f is invertible if and only if f is both injective (i.e., one-to-one) and surjective (i.e., onto), (b) if f is invertible then it has a unique inverse.

Note that being an *if and only if* statement, part (a) requires two arguments. One of the two is so immediate that it may be confusing to articulate it precisely. Try to do the rest too as concisely and incisively as you can. It may help to explore the consequence of just one of the two required equations and prove a subclaim. Does $f \circ g = id_T$ imply something about f ? About g ?

Optional extensions. (c) *Results mining the same vein.* A function $f : S \rightarrow T$ is called left invertible if it has a left inverse, i.e., a function $g : T \rightarrow S$ with $gf = id_S$. (We’re dropping $\circ$ from now on.) Which functions have a left inverse? If a left inverse exists, must it be unique? Repeat for right inverses. What can you say about a function having a left inverse and a right inverse?

Now call f right cancellable if for any functions h_1, h_2 , the equation $h_1f = h_2f$ implies $h_1 = h_2$. Can you see an immediate formal connection between being left/right cancellable and left/right invertible? Which functions are right cancellable? Left cancellable? Both?

Note: This part gets one’s toes wet in formulating statements purely in terms of functions and their composition as opposed to using elements. Analogues of such statements and this style of doing business *might* become relevant for some of you later on when you deal with categories of objects other than plain sets. Unless and until that happens, don’t get too fond of such things.

(d) *Two-out-of-three property.* See informally for yourself that any two of the following properties for a function between *finite* sets implies the third: injective, surjective, domain and codomain have the same finite cardinality. In particular a function between finite sets of equal cardinality is injective if and only if it is surjective. This is not so exciting for sets but its analogue for vector spaces is quite useful. What happens if we drop the word *finite* everywhere?

2. Consider the commonly made statement “Defining an equivalence relation on a set S is the same as defining a partition of S ”. Make the preceding vague statement precise. Then prove the precise statement. Both underlined notions are defined below. Hint: I almost made this a part of the previous problem.

A *partition* of S is defined to be a set of pairwise disjoint subsets of S whose union is S , i.e., a partition of S is a set $\{S_\alpha | \alpha \in I\}$, where α ranges over some index set I and each $S_\alpha \subset S$, such that $\bigcup_{\alpha \in I} S_\alpha = S$ and for all distinct $\alpha, \beta \in I$, one has $S_\alpha \cap S_\beta = \emptyset$.

Recall that a relation on a set S means a relation from S to itself, i.e., a subset of $S \times S$. A relation $\sim$ on a set S is called an equivalence relation if it is

- reflexive (i.e., for each $x \in S$, one has $x \sim x$),
- symmetric (i.e., whenever $x \sim y$, one must also have $y \sim x$), and
- transitive (i.e., whenever $x \sim y$ and $y \sim z$, one must also have $x \sim z$).

As an aside, recall that changing *symmetric* to *antisymmetric* gives the notion of a partial order, a completely different kind of beast.

3. Let $f : S \rightarrow T$ be a function. Let us make two definitions, which might be familiar from school: for $A \subset S$, let $f(A) = \{f(x) | x \in A\}$, a subset of T . For $B \subset T$, let $f^{-1}(B) = \{x | f(x) \in B\}$, a subset of S . The latter notation does not require f^{-1} to be a function from T (but can be thought of as defining a function from the power set of T to that of S). [Answer the following with proofs/counterexamples as necessary.](#) While all questions are phrased as asking for a YES/NO answer, supply the strongest results that you can for all parts.

- (a) Is $f(A_1 \cup A_2) = f(A_1) \cup f(A_2)$, i.e., does f preserve finite unions? (Why “i.e.”?) Arbitrary unions?
- (b) Does f preserve finite intersections? Arbitrary intersections?
- (c) Does f preserve complements, i.e., is $f(A^c) = f(A)^c$? More precisely, is $f(S \setminus A) = T \setminus f(A)$?
- (d) Repeat the previous questions for f^{-1} .
- (e) Is $f^{-1}(f(A)) = A$? Is $f(f^{-1}(B)) = B$?
- (f) For each NO answer, does it change to YES for some functions f other than isomorphisms?

4. Here is a small exercise in set theory language to work out. A “family of subsets of a set S indexed by a set I ” means, formally speaking, a function from I to the set of subsets of S . In this context the subset of S that is the image of $i \in I$ is simply denoted S_i .

- (a) Define $\bigcup_{i \in I} S_i = \{x \in S | \exists i \in I \text{ with } x \in S_i\}$. What is the meaning of this when I is the empty set?
- (b) Similarly define $\bigcap_{i \in I} S_i$. Deduce its meaning when I is the empty set.

Regarding basic set theory. Your CMI courses are unlikely to discuss basic set theory in class for its own sake. Look at any suitable text (examples below) to confirm your familiarity with basics such as interrelationships among union, intersection and complement. You do need to know these well in order to use as required.

Additionally, when you need it, you should also learn the meaning of countability and uncountability and basic facts about these notions, which are very interesting and of major historical/foundational significance. E.g., the set of rational numbers is countable, i.e., “there are as many rational numbers as there are integers” (what??), but the set of real numbers is uncountable. So there are strictly fewer rational numbers than there are real numbers. (Wait, why is this not obvious?!) References: many, including good analysis textbooks, e.g., Bartle-Sherbert (chapter 1) or Tao’s Analysis 1 (chapters 3, 8) or baby Rudin (2.1 to 2.14).